Introduction

For this analysis, we use EPA EJScreen 2024 (v2.32), with diesel PM estimates from EPA's 2020 AirToxScreen model, to examine how demographic composition, population density, income, geographic location, and traffic proximity relate to diesel particulate matter concentrations across U.S. census tracts.

The first part of the analysis uses regression models to examine how diesel PM concentrations vary with the percentage of residents of color, while accounting for population density, income, and regional differences. The second part focuses on traffic proximity, examining its relationship with diesel PM and whether including it in the models changes the observed association between demographic composition and pollution levels.

Population density, income, and geographic location and Diesel Particulate Matter

This section asks: residents of different demographic groups live with different levels of modelled diesel particulate matter (PM). How much of that gap remains once we account for population density, income, and geographic location? **Table 1** lists the models tested. All of the models use the same 84,222 census tracts. The outcome is log10 diesel PM, so coefficients are reported as percent changes.

Model	**Type**	**Formula (outcome ~ predictors)**	**What it asks**
m1	Simple linear	log_dsl ~ poc10	Do tracts with more residents of color have more diesel PM?
m2	Multiple linear	log_dsl ~ poc10 + log_density	Among tracts with the same density, do those with more residents of color have more diesel PM?
m3	Multiple linear	log_dsl ~ poc10 + log_density + lowinc10	Among tracts with the same density and income, do those with more residents of color have more diesel PM?
m4	Multiple linear (main)	log_dsl ~ poc10 + log_density + lowinc10 + region	Is the relationship still present when comparing tracts in the same EPA region?
m4c	Multiple linear	m4 with centered	What does a typical tract look like?

		predictors	
m4_rl	Multiple linear	m4 with Southeast as reference	Does the choice of reference region change the interpretation?
m5	Multiple linear + interaction	log_dsl ~ poc10 * region + log_density + lowinc10	Does the relationship between residents of color and diesel PM differ across regions?

Table 1. Outline of all the models used in this portion of the project.

Simple regression

In Model 1, each additional 10 percentage points of residents of color was associated with 10.6% more modelled diesel PM (95% CI: 10.4% to 10.8%; **Figure 1**). A tract with no residents of color is predicted to have 0.097 µg/m³. Race alone explained 16% of the variation (R^2 = 0.158), and a typical prediction was off by a factor of about 2.

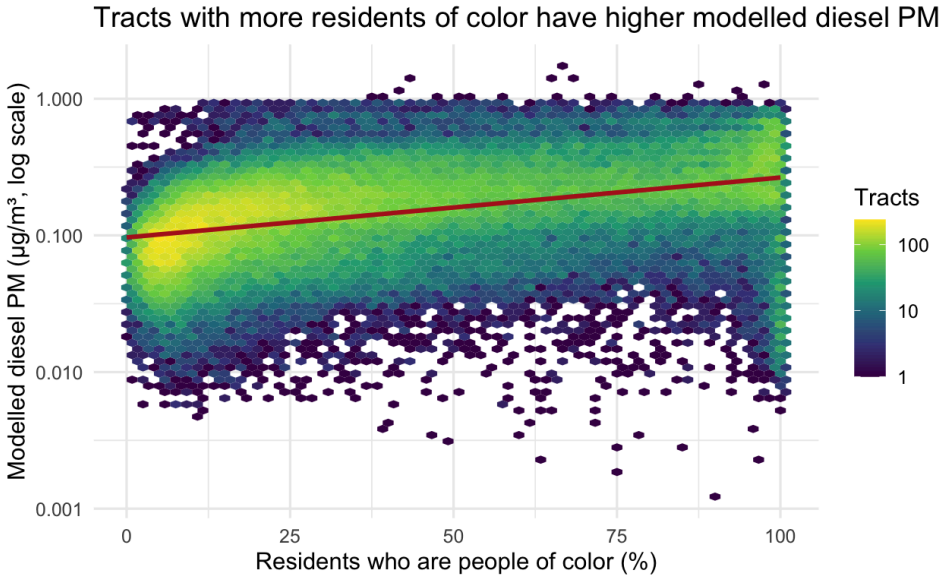


Figure 1. Each hexagon represents a tract (84,222) and the line is the simple regression with its 95% confidence interval band. Diesel PM is on a log scale.

Multiple regression

Models 2–4 added density, income, and EPA region one at a time (**Table 2**). Density reduced the race coefficient to 0.8% (CI: −0.4% to 2.0%) and raised R^2 from 0.158 to 0.614, so density accounts for most of the simple association. Adding income raised the coefficient to 2.5%, a sign of suppression: tracts with more residents of color tend to be lower-income, and lower-income tracts have slightly less diesel PM at the same density. With region added, the coefficient was 2.9% (CI: 1.9% to 3.8%).

model <chr>	poc_pct <dbl>	clustered_low <dbl>	clustered_high <dbl>	r_squared <dbl>	adj_r_squared <dbl>	rse <dbl>
1. People of color only	10.6	8.7	12.6	0.158	0.158	0.304
2. + density	0.8	-0.4	2	0.614	0.614	0.205
3. + income	2.5	1.2	3.9	0.625	0.625	0.202
4. + EPA region	2.9	1.9	3.8	0.642	0.642	0.198

Table 2. R^2 and RSE values adding each individual predictor for each model explored.

In the full model (**Table 3**), holding the other predictors constant, a tract ten times denser had 84.2% more diesel PM, and 10 more points of low-income residents went with 5.1% less. Every region had more diesel PM than New England, the reference level, from 9.5% in Mountains & Plains (CI includes zero) to 50.5% in the Pacific Northwest. The predictors were not too correlated to separate (VIFs 1.38 to 2.01, all below 5). Centering the predictors changed the intercept from an unrealistic 0.023 μg/m³ to a realistic 0.113 μg/m³ for a typical New England tract, without changing any slope. Changing the reference region to the Southeast changed only the region comparisons and the intercept.

term <chr>	estimate <dbl>	pct_change <dbl>	clustered_low <dbl>	clustered_high <dbl>	t_value <dbl>	p_value <dbl>
(Intercept)	-1.6396	-97.7	-97.9	-97.5	-74.47	0.00E+00
poc10	0.0123	2.9	1.9	3.8	5.87	4.28E-09
log_density	0.2652	84.2	79.4	89.1	45.68	0.00E+00
lowinc10	-0.0229	-5.1	-6.2	-4.1	-9.27	1.97E-20
region2 NY/NJ/PR	0.0801	20.2	1.6	42.3	2.15	3.18E-02
region3 Mid-Atlantic	0.1083	28.3	18.6	38.8	6.22	5.12E-10
region4 Southeast	0.1106	29	19.1	39.7	6.27	3.60E-10
region5 Great Lakes	0.1592	44.3	31.7	58.1	7.84	4.45E-15
region6 South Central	0.0698	17.4	6.7	29.2	3.29	9.96E-04
region7 Central Plains	0.166	46.5	35.2	58.9	9.27	1.84E-20
region8 Mountains & Plains	0.0396	9.5	-5.6	27.1	1.2	2.28E-01
region9 Pacific Southwest	0.1356	36.7	15.3	62	3.6	3.14E-04
region10 Pacific Northwest	0.1777	50.5	27.3	78	4.79	1.68E-06

Table 3. These outputs are all of the estimates on a log10 scale for each of the models analyzed in this section of the project.

Geography

Letting the race slope differ by region improved the model, though adjusted R^2 rose only from 0.642 to 0.650. Fit separately within each region, the association was positive in 7 of 10 regions (**Figure 2**). It was strongest in Mountains & Plains (9.0%) and the Pacific Northwest (8.5%), about three times the national 2.9%, and essentially zero in NY/NJ/PR and South Central. The national estimate is a compromise that hides where disparities are steepest.

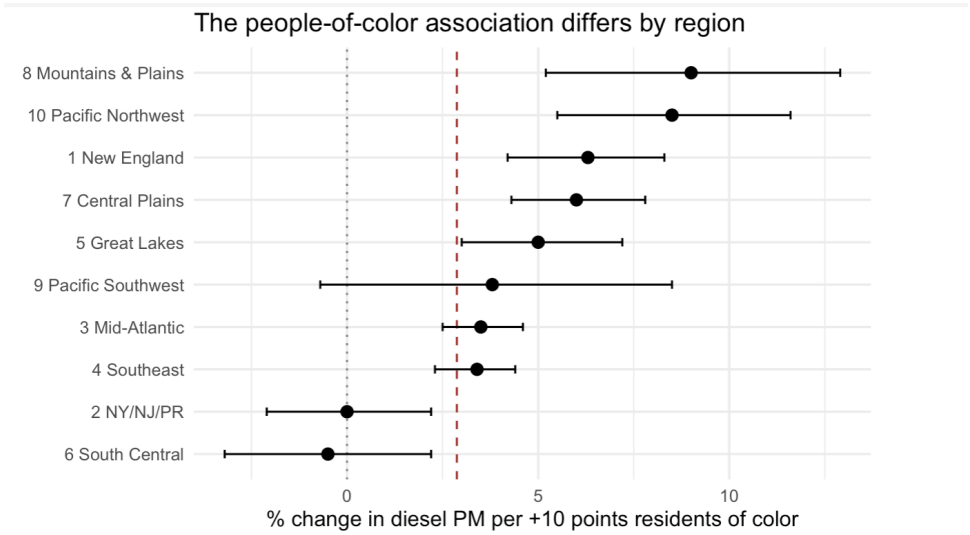


Figure 2. Percent change in modelled diesel PM per 10 more percentage points of people of color estimated separately for each EPA region, holding density and income constant. Bars are 95% county clustered confidence intervals. The dashed line is the national estimate (2.9%) and the dotted line is zero.

Diagnostics

The residual plots support a linear model (**Figure 3**). The Q-Q plot showed heavy tails, and the spread grew slightly at high predicted values, but with 84,222 tracts neither has much effect on the coefficients. Residuals were centred near zero in every region, though the fit was looser in regions that mix dense cities with remote areas (**Figure 4**). Independence fails because neighbouring tracts share highways and air-model inputs, so we clustered standard errors by county. This widened the race coefficient's CI about six-fold, from 2.7%–3.0% to 1.9%–3.8%.

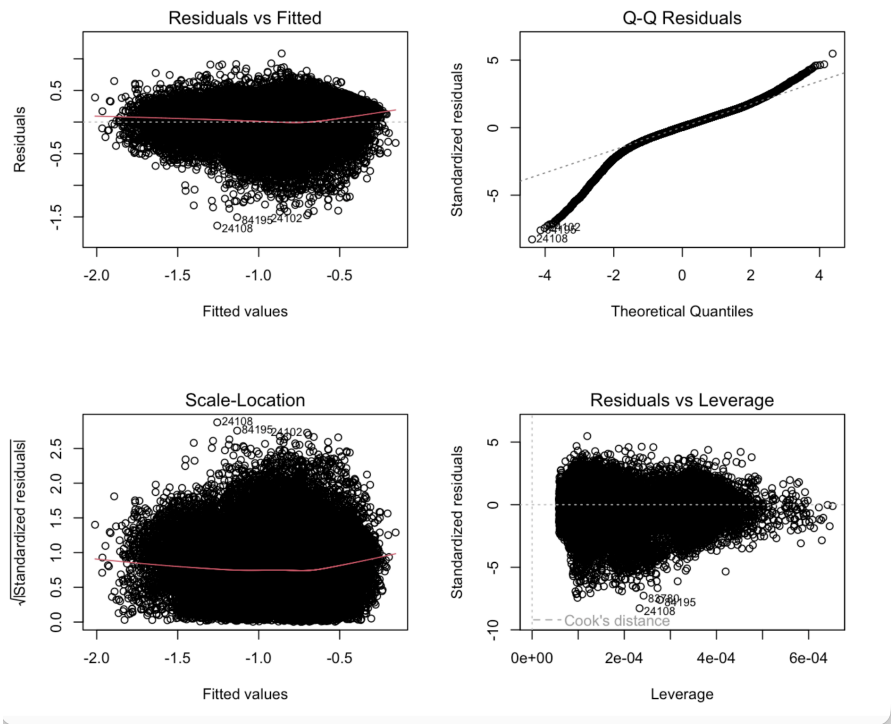


Figure 3. Standard diagnostic plots for the full model (m4).

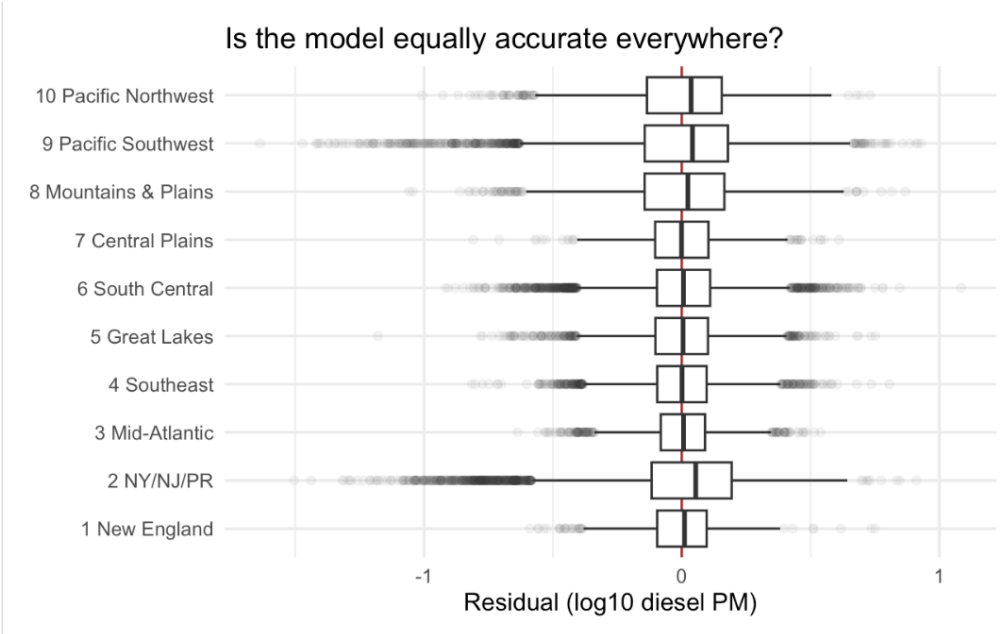


Figure 4. Residuals (actual minus predicted log10 diesel PM) from the full model, by EPA region. Every box is centered near zero, so the model is not systematically too high or too low in any region. Wider boxes in NY/NJ/PR, Mountains & Plains, and the Pacific Southwest and Northwest mean the model fits those regions less precisely.

Effect sizes and testing

On a standardized scale, density's effect (β = 0.739) was about 6.7 times that of race (β = 0.111), and income was similar in size to race but negative (β = −0.126). By Cohen's f^2, density was large (1.046), and region was second (0.046), ahead of income (0.032) and race (0.017). In real units, moving from the 25th to the 75th percentile of residents of color predicts 15% more diesel PM after adjustment (CI: 10% to 21%). The race coefficient sits 5.87 clustered standard errors from zero ($p = 4.3 \times 10^{-9}$), and the study could detect effects as small as 1.36%, so we rely on effect sizes and confidence intervals rather than p-values.

Logistic regression

Using EPA's 80th-percentile cutoff (19.9% of tracts flagged), each additional 10 points of residents of color was associated with 18% higher odds of a tract being in the national top 20% for diesel PM (OR = 1.18, CI: 1.11 to 1.26). A tenfold increase in density multiplied the odds by 20 (CI: 14.2 to 28.2), and more low-income residents lowered them (OR = 0.89, CI: 0.82 to 0.96). Compared with New England, the Pacific Northwest had the highest odds (OR = 17.5, CI: 6.4 to 47.7). The Mid-Atlantic, South Central, and Central Plains could not be distinguished from New England. The model was well calibrated, but the log-odds curve flattens at high shares of residents of color, so the odds ratio of 1.18 is an average. **Figure 5** shows the final geographic model against the data in each region.

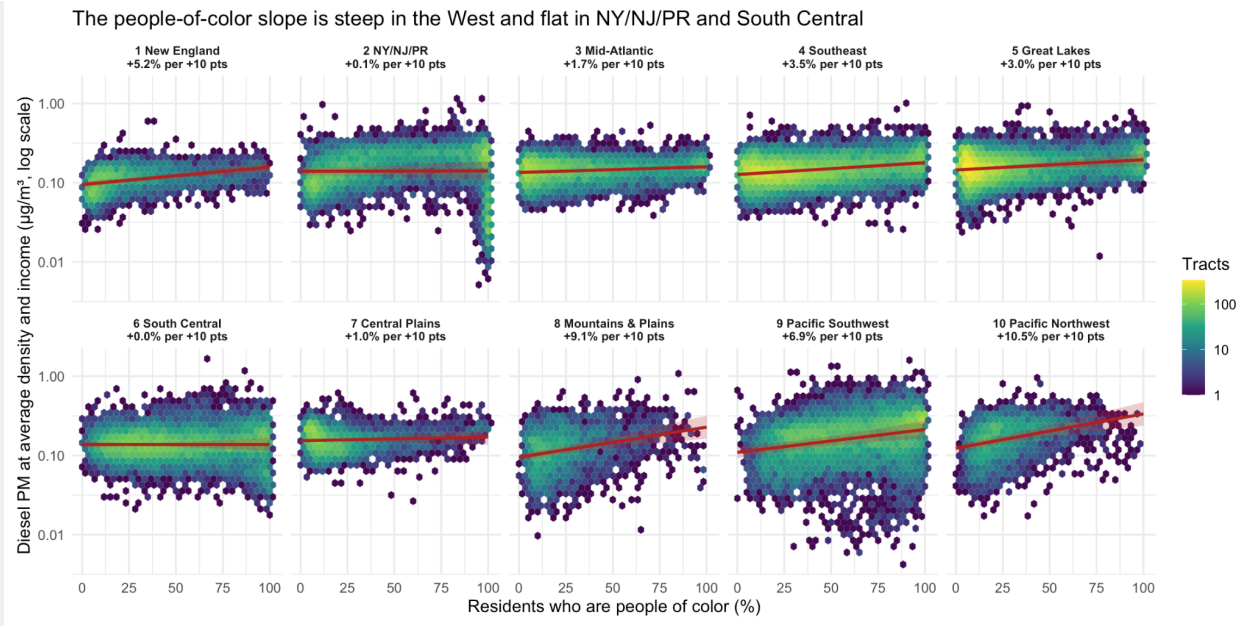


Figure 5. The final geographic model (m5) against the data. Each panel is one EPA region. The red line is m5's predicted diesel PM at that average density and income, with a 95% county-clustered confidence band.

W.E.B. Du Bois Figure

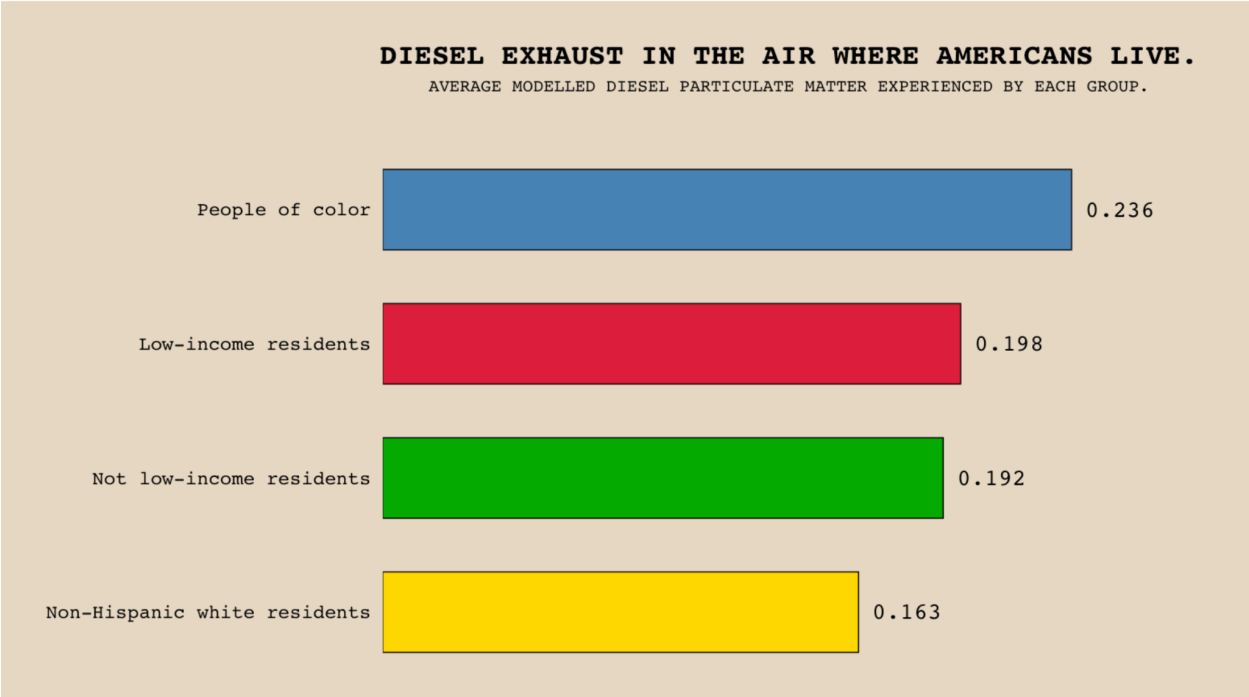


Figure 6. Population-weighted modelled diesel PM by group. Each bar is the average level the typical member of the group lives with in

Traffic Proximity and Diesel Particulate Matter

To better understand the relationship between transportation-related conditions and environmental disparities, we extended our demographic analysis using EPA EJScreen 2024 data. Our analysis focused on two research questions:

RQ1: How strongly is traffic proximity associated with modeled diesel particulate matter (PM) concentrations across U.S. census tracts?

RQ2: Does accounting for traffic proximity change the association between the share of residents of color and modeled diesel PM, after controlling for population density, low-income share, and EPA region?

We analyzed 84,222 census tracts with valid demographic, geographic, traffic proximity, and diesel PM data. We used simple linear regression to address RQ1 and compared two multiple linear regression models to address RQ2. Traffic proximity

and population density were transformed using log(1 + x) to reduce skewness, while the multiple regression models used log10 diesel PM as the outcome.

Simple linear regression

The simple model used diesel PM in its original units (µg/m³) and log-transformed traffic proximity as the predictor.

Table 1. Simple linear regression estimates

Predictor	Coefficient (β)	95% CI
Intercept	−0.3656	[−0.3703, −0.3608]
Traffic proximity, log(1 + x)	0.0422	[0.0418, 0.0426]

Outcome: modeled diesel PM (µg/m³). N = 84,222; R^2 = 0.3924. Intervals are conventional OLS confidence intervals.

A one-unit increase in log-transformed traffic proximity was associated with 0.0422 µg/m³ higher modeled diesel PM (95% CI: 0.0418–0.0426). Traffic proximity alone explained 39.2% of the variation in diesel PM across tracts.

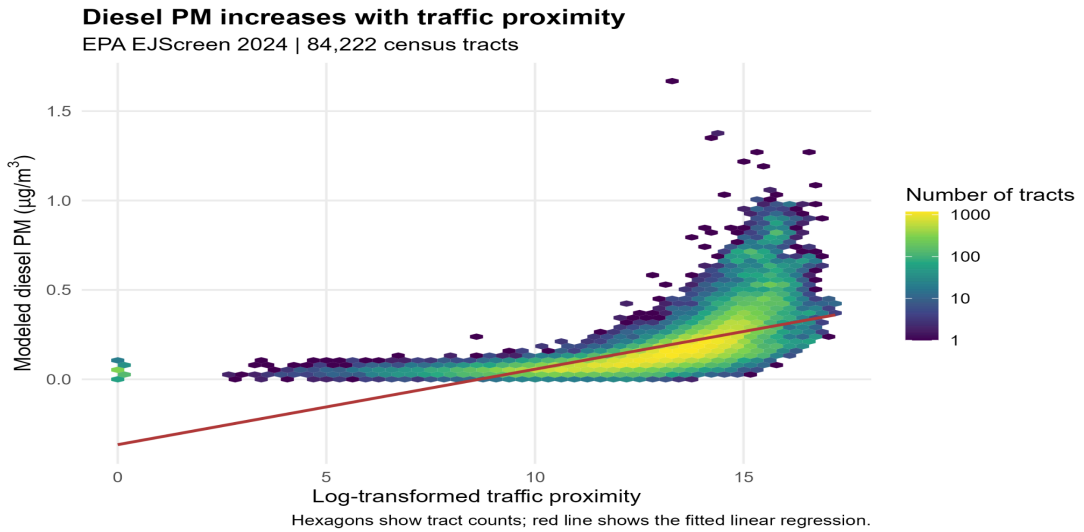


Figure 7. Relationship between log-transformed traffic proximity and modeled diesel PM. Hexagons represent groups of census tracts, and the fitted line summarizes the unadjusted association.

The positive trend is clear, although diesel PM still varies substantially among tracts with similar traffic proximity values. This suggests that traffic is relevant to the observed pattern but does not capture all the conditions associated with diesel pollution.

Multiple linear regression

We compared two multiple linear regression models using log10 diesel PM as the outcome. Both used the same 84,222 census tracts and included the share of residents of color, population density, low-income share, and EPA region. Traffic proximity was added to the second model to assess its contribution beyond these factors.

Table 2. Multiple linear regression coefficients and 95% confidence intervals

Predictor	**Without traffic: β [95% CI]**	**With traffic: β [95% CI]**
Intercept	−1.7622 [−1.7707, −1.7538]	−2.2292 [−2.2408, −2.2175]
Share of residents of color	0.1193 [0.1130, 0.1257]	0.0674 [0.0614, 0.0734]
Traffic proximity, log(1 + x)	—	0.0543 [0.0533, 0.0553]
Population density, log(1 + x)	0.1170 [0.1163, 0.1178]	0.0768 [0.0757, 0.0778]
Low-income share	−0.2297 [−0.2383, −0.2210]	−0.1553 [−0.1635, −0.1471]
Region 2: NY/NJ/PR	0.0792 [0.0714, 0.0870]	0.1329 [0.1256, 0.1403]
Region 3: Mid-Atlantic	0.1087 [0.1009, 0.1164]	0.1525 [0.1452, 0.1598]
Region 4: Southeast	0.1121 [0.1050, 0.1193]	0.1584 [0.1516, 0.1651]
Region 5: Great Lakes	0.1596 [0.1524, 0.1668]	0.1883 [0.1815, 0.1951]

Region 6: South Central	0.0702 [0.0626, 0.0778]	0.1258 [0.1186, 0.1330]
Region 7: Central Plains	0.1656 [0.1566, 0.1745]	0.1954 [0.1870, 0.2038]
Region 8: Mountains & Plains	0.0347 [0.0252, 0.0441]	0.0646 [0.0556, 0.0735]
Region 9: Pacific Southwest	0.1348 [0.1273, 0.1423]	0.1561 [0.1490, 0.1632]
Region 10: Pacific Northwest	0.1764 [0.1671, 0.1858]	0.2131 [0.2043, 0.2219]

Outcome: log10 modeled diesel PM. N = 84,222 for each model. EPA Region 1 (New England) is the reference category. The share variables are measured from 0 to 1. Confidence intervals are conventional OLS intervals, not county-clustered.

The demographic coefficient decreased from 0.1193 to 0.0674 after including traffic proximity. This corresponds to a reduction in the estimated diesel PM difference from 2.79% to 1.56% for a 10-percentage-point increase in the share of residents of color, holding the other variables constant.

Traffic proximity remained positively associated with modeled diesel PM in the fully adjusted model (β = 0.0543, 95% CI: 0.0533–0.0553). A one-unit increase in log-transformed traffic proximity was associated with approximately 13.3% higher modeled diesel PM. Population density also remained positively associated with diesel PM, while low-income share had a negative conditional association. Regional coefficients varied relative to New England, indicating that geographic differences remained after adjustment.

Table 3. Model comparison and interpretation

Measure	**Without traffic**	**With traffic**
Census tracts	84,222	84,222
R^2	0.6422	0.6848

Share of residents of color, β	0.1193	0.0674
Diesel PM difference per 10 percentage points more residents of color	+2.79%	+1.56%
Traffic proximity, β	—	0.0543

Percentage differences are derived from log10 regression coefficients. The two adjusted models use identical observations and outcome transformations.

Adding traffic proximity increased R^2 by 0.043, meaning the model explained approximately 4.3 additional percentage points of variation in log10 diesel PM. The demographic coefficient decreased by approximately 44%. This suggests that traffic proximity accounts statistically for part of the demographic association, although the coefficient change does not establish a causal mediation effect.

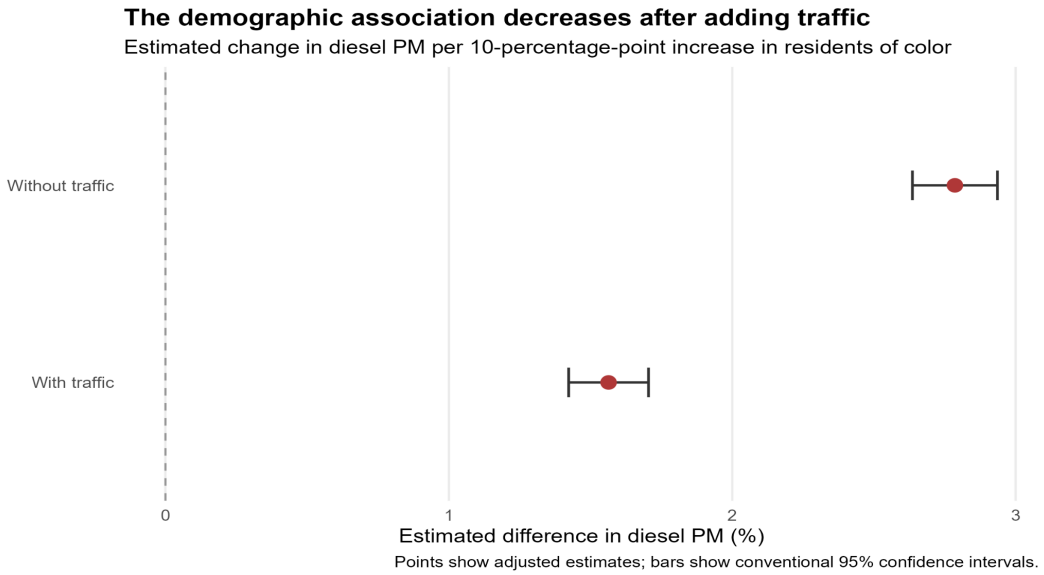


Figure 8. Estimated percentage difference in modeled diesel PM associated with a 10-percentage-point increase in the share of residents of color, before and after accounting for traffic proximity. Horizontal bars represent conventional 95% confidence intervals.

Variation across traffic-proximity levels

We divided census tracts into ten equally sized groups based on traffic proximity. Mean modeled diesel PM increased from 0.053 µg/m³ in the lowest decile to 0.427 µg/m³ in the highest, an approximately 8.1-fold difference.

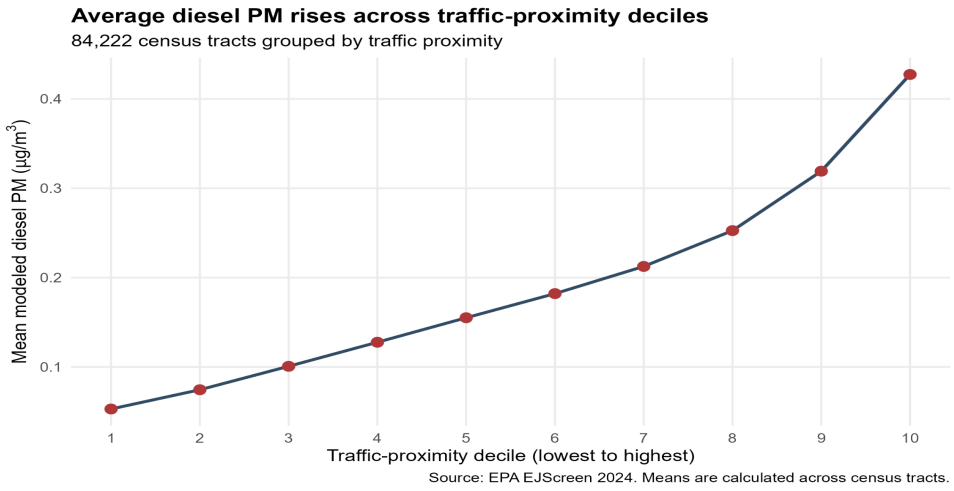


Figure 9. Mean modeled diesel PM across traffic-proximity deciles, from lowest to highest. Each point represents an unweighted census-tract average. Source: EPA EJScreen 2024.

The decile pattern is consistent with the regression findings, but the averages are unadjusted and may also reflect differences in demographic composition, population density, and regional characteristics.

Regression diagnostics

Residual diagnostics were examined for all three regression models. In the fully adjusted model, the residuals-versus-fitted plot showed some curvature, suggesting that the linear specification may not capture every pattern in the data. The scale-location plot indicated unequal residual variance, while the Q-Q plot showed departures from normality in the tails. Cook's distance identified potentially influential observations, but no single tract appeared to dominate the fitted model visually.

Independence is another concern. Neighboring tracts may share highways, pollution sources, and environmental conditions, introducing spatial correlation. The models use conventional standard errors, which may produce confidence intervals that are too narrow when residual variance is unequal or nearby

observations are correlated. These issues limit the precision of statistical inference, even when the estimated associations are substantial.

Measurement considerations and limitations

EJScreen diesel PM concentrations are modeled environmental estimates rather than direct measurements of individual exposure. Traffic proximity is also a constructed indicator, not a direct measure of diesel emissions or the amount of pollution an individual inhales (U.S. EPA, 2024).

Both measures operate at the census-tract level. Residents within the same tract may experience different pollution conditions depending on their distance from major roads, workplace locations, or daily travel patterns. The analysis also weights each tract equally, regardless of population, and excludes tracts with fewer than 100 residents or missing required data. Consequently, the results describe relationships across the included tracts rather than average exposure among all individuals.

Other factors, including industrial activity, freight movement, and local land use, may contribute to modeled diesel PM. The analysis is observational, so these factors and possible measurement errors limit causal interpretation.

Interpretation

The results indicate that traffic proximity is closely associated with modeled diesel PM and helps explain part of the demographic differences observed in the broader analysis. The simple regression explained 39.2% of variation in diesel PM, while adding traffic proximity to the multiple regression increased R^2 from 0.642 to 0.685.

The key finding is the reduction in the demographic association after including traffic. A 10-percentage-point higher share of residents of color was associated with 2.79% higher modeled diesel PM before accounting for traffic and 1.56% afterward. This suggests that traffic-related conditions are associated with part of the demographic difference, but the remaining association indicates that the included variables do not fully account for it.

These findings should not be interpreted as evidence that demographic composition causes pollution or that traffic proximity directly mediates the disparity. The models describe associations between neighborhood characteristics

and modeled pollution, not individual exposure or the effects of an intervention. A claim that residents of color personally inhale 1.56% more diesel pollution would therefore go beyond what these data can establish.

Overall, the traffic analysis adds a transportation-related dimension to the environmental justice findings. It shows that understanding differences in modeled diesel PM requires attention not only to demographic composition but also to the environmental and geographic conditions associated with where people live.

Acknowledgements

Manasvi did the traffic proximity analysis and Kayla did the EPA region analysis.

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